

- 10 (a) The photoactivation of chlorophyll is an essential process in the conversion of light energy to chemical energy during the light-dependent stage of photosynthesis.

Photoactivation of chlorophyll leads to the synthesis of ATP.

Outline how the photoactivation of chlorophyll leads to the production of ATP.

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- (b) (i) ATP produced in the light-dependent stage passes to the Calvin cycle of the light-independent stage.

Name **two** compounds in the Calvin cycle that are formed using this ATP. Do **not** give abbreviations for these compounds.

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[2]

- (ii) Reduced NADP is also produced in the light-dependent stage.

Outline the role of reduced NADP in the light-independent stage.

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- (iii) Some glycerate 3-phosphate (GP) molecules produced in the Calvin cycle are used to produce a type of molecule that is **not** an intermediate in the Calvin cycle.

Name the type of molecule that can be produced from GP that is **not** an intermediate in the Calvin cycle.

..... [1]

- (c) Environmental factors affect the rate of photosynthesis.

Figure 10.1 shows the effect of carbon dioxide concentration on the rate of photosynthesis.

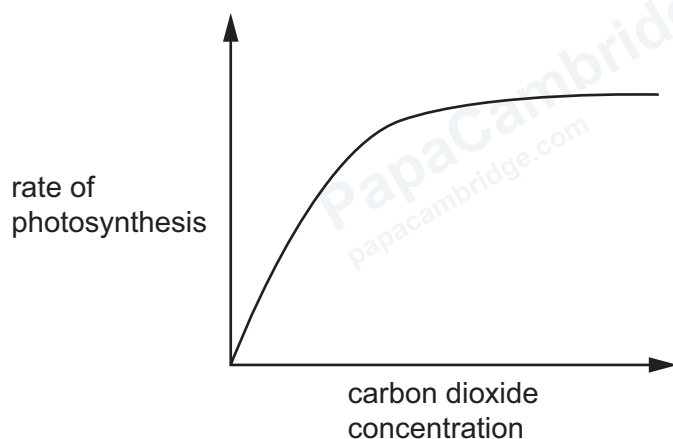


Figure 10.1

Explain the reasons for the relationship shown in Figure 10.1.

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[Total: 12]